

Function Reference & Interview Prep

A concise index of all key classes and their methods, with one-line descriptions.

1. FormGame (UI)

- `FormGame(SqlConnection, string)`: constructor initializes DB connection, player name, and managers.
- `InitializeManagers()`: sets up `GameBoardManager`, `GameLogicManager`, and `GameFlowManager`.
- `EnsureBoardReady()`: builds or resets the board UI, shows controls, and starts turn tracking.
- `OnCellClick(object, EventArgs)`: handles clicks to select pieces or perform moves.
- `IsSelectingPiece(int, int)`: returns true if a board cell contains the current player's piece.
- `IsChoosingMove(int, int)`: returns true if a board cell is a highlighted valid move.
- `UpdateTurnLabel()`: updates the current-turn label text.
- `radioSize_CheckedChanged(...)`: reinitializes managers and board on size change.
- `btn_twoPlayers(...)` & `btn_againstCom(...)`: set up game modes.
- `btn_changeColor(...)`: opens color picker and updates highlight color.
- `btn_gameRoles(...)`: displays game rules.
- `timer1_Tick(...)`: decorative LED toggling using `SendKeys`.
- `StartGameMode(bool)`: locks mode choice and displays scores.
- `ShowPlayerScore(string)`: retrieves and shows a player's score.
- `LoginSecondPlayer()`: shows login form for second player and updates state.

2. GameBoardManager

- `GameBoardManager(Panel, int, Color, Action)`: constructor stores dependencies.
- `GetBoardButtons()`, `GetBoardLogic()`: expose internal arrays.
- `BuildBoard(EventHandler)`: creates button grid, sizes and attaches click handlers.
- `ResetBoard(string, string, string)`: clears board and places starting pieces.
- `HighlightAvailableMoves(List<Move>)`: colors valid move destinations.
- `ResetMoveHighlights()`: resets highlight colors.
- `PickHighlightColor(Button)`: shows color dialog, updates highlight color and button forecolors.
- `GetHighlightColor()`: returns the current highlight color.
- `UpdateBoardLayout()`: recalculates button sizes and positions on panel resize.

3. GameLogicManager

- `GameLogicManager(int)`: constructor stores board dimension.
- `List<Move> GetMoves(int[,], int, int, bool)`: returns all legal moves (normal and capture) with heuristic scores.
- `bool CheckWin(int[,], bool)`: checks if opponent has no moves left.
- Helpers: `InBounds`, `SimulateMove`, `SimulateCapture` for move simulation.

4. GameFlowManager

- `GameFlowManager(...)`: constructor wires up board, UI, logic, database, AI, and callbacks.
- `ExecutePlayerMove(int, int, int, int)`: applies a human move, captures, promotions, then calls `CompleteTurn`.
- `ExecuteComputerMove()`: selects and applies an AI move, then calls `CompleteTurn(false)`.

4. GameFlowManager (continued)

- CompleteTurn(bool): checks for win, toggles turn, resets highlights, updates labels.
- HandleWin(bool): updates scores, records game, shows message, and invokes end-of-game callback.
- static DataTable ExecuteSelectQuery(string, SqlConnection): helper for raw SELECT queries.
- AnalyzeComputerMoves(): (private) classifies moves into winning, threats, and risky.

5. FormPlayer

- FormPlayer(SqlConnection): constructor configures chart and loads player grid.
- LoadAllPlayerStats(): binds all player stats to DataGridView.
- dg_Players_CellMouseClicked(...): shows pie chart of win vs. loss percentages.

6. FormLogIn

- FormLogIn(SqlConnection): constructor initializes repository.
- btn_LogIn_Click(...): validates username/password and sets Player property.
- btn_Forgot_Click(...): opens FormForgotPassword.
- btn_NewUser_Click(...): opens FormNewUser.

7. FormResetPassword

- FormResetPassword(): constructor masks password fields.
- btnOk_Click(...): validates non-empty, minimum length, and match before setting NewPassword.
- btnCancel_Click(...): cancels dialog.

8. PasswordHelper

- CreateHash(string, out string, out string): generates random salt and PBKDF2-SHA256 hash.
- VerifyPassword(string, string, string): recomputes hash and compares to stored.
- HexStringToBytes(string): converts hex string to byte array.

9. SqlGameResultRepository

- EnsureComputerExists(): inserts default computer user if missing.
- UsernameExists(string): checks for user existence.
- CreatePlayer(...): inserts new user with 'salt:hash' password.
- ValidatePassword(...): verifies legacy plaintext or salted hash, migrates plaintext.
- ChangePassword(...): updates user's password to new migrated format.
- GetPassword, GetPrivateQuestion, GetAnswer: retrieve single fields.
- GetPrivateQuestionTable(...): returns DataTable for UI binding.
- UpdateScore(string,int): increments scoring.
- RecordWin(...): logs completed game.
- GetScore, GetAllUsernames, GetAllPlayerStats, GetPlayerStats, GetAllGames, GetGameByNumber: various queries.
- DeleteGame(...): admin-only deletion and scoreboard refresh.
- RefreshPlayerScores(): recomputes win/lose counts and overall scoring.
- LogError(Exception): logs exceptions.
- Helpers: ExecuteNonQuery, QuerySingle<T>, ExecuteSelect.